

Deep Dive: 14. Error Handling

Concept Summary

Error handling prevents your program from crashing by 'catching' exceptions gracefully.

Coding Structure

```
try:
    # risky code
except ErrorType:
    # fallback code
```

Example Code

```
try:
    x = 10 / 0
except ZeroDivisionError:
    print('Cannot divide by zero!')
finally:
    print('Execution complete.')
```

Code Explanation

- Line 1-2: Python attempts the risky code (dividing by zero).
- Line 3-4: Since line 2 causes a ZeroDivisionError, Python instantly jumps here instead of crashing the program.
- Line 5-6: The 'finally' block ALWAYS runs, whether an error happened or not. Great for cleanup tasks.

Practice Problems

1. Write a try/except block that catches a ValueError when trying to convert int('hello').
2. Ask the user for a number. Use try/except to keep asking until they provide a valid integer.